Video Object Tracking on MOT16 Using Mask R-CNN and Siamese Re-Identification

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Abstract

Multi-Object Tracking and Segmentation (MOTS) is a fundamental task in computer vision with applications spanning autonomous driving, surveillance, sports analytics, and robotics. This project presents a complete end-to-end pipeline for multi-pedestrian tracking on the MOT16 benchmark. The system combines three core components: (1) a **Mask R-CNN** detector fine-tuned on MOT16 for person detection, (2) an **improved Siamese ReID network** trained on Market-1501 for appearance-based re-identification, and (3) a **DeepSORT-style tracker** using the Hungarian algorithm with a combined IoU and cosine-distance cost matrix for cross-frame identity association. The ReID model achieves **73.93% mAP** and **89.22% Rank-1 accuracy** on Market-1501. The integrated pipeline successfully tracks **22 unique identities** across **525 frames** on MOT16-09.

1 Introduction

1.1 Problem Statement

Video object tracking requires detecting and following multiple objects across consecutive video frames while maintaining consistent identity assignments. This is particularly challenging due to occlusions, appearance changes, scale variation, and crowded scenes where pedestrians frequently overlap.

1.2 Approach Overview

Our approach follows an offline tracking paradigm consisting of three stages:

1. **Detection:** Each frame is processed by a fine-tuned Mask R-CNN to produce person bounding boxes with confidence scores.
2. **Re-Identification (ReID):** Detected person crops are fed through a Siamese network to extract 256-dimensional appearance embeddings.
3. **Tracking:** A DeepSORT-style tracker associates detections across frames using the Hungarian algorithm with a composite cost function balancing spatial (IoU) and appearance (cosine distance) cues.

The pipeline is summarized as:

`Frame` → `Detect` → `Crop` → `ReID Embed` → `Hungarian Match` → `Track` → `Video`

1.3 Datasets

- **MOT16** (Multi-Object Tracking Benchmark 2016): Used for detector fine-tuning and tracking evaluation. Contains 7 training and 7 test sequences with ground truth in the format `(frame, id, bb_left, bb_top, bb_width, bb_height, conf, class, visibility)`.
- **Market-1501**: Used for ReID training. Contains 32,668 images of 1,501 identities from 6 cameras, split into 12,936 training images (751 IDs), 3,368 query images, and 19,732 gallery images.

2 Methodology

2.1 Person Detection — Fine-Tuned Mask R-CNN

2.1.1 Architecture

We employ **Mask R-CNN** with a **ResNet-50-FPN** backbone, initialized from COCO-pretrained weights. The detection head is replaced with a custom `FastRCNNPredictor` for **2 classes** (background + person).

```
model = torchvision.models.detection.maskrcnn_resnet50_fpn(
    weights='DEFAULT', min_size=384, max_size=640)
in_features = model.roi_heads.box_predictor.cls_score.in_features
model.roi_heads.box_predictor = FastRCNNPredictor(in_features, 2)
```

Key configuration: input min_size=384, max_size=640; detection threshold=0.5.

2.1.2 Training Details

- **Optimizer:** AdamW with differential learning rates (Heads: 3×10^{-3}, Backbone: 3×10^{-4}), weight decay=1×10^{-4}
- **LR Schedule:** Linear warmup (4 epochs) + Cosine annealing
- **Epochs:** 40 with early stopping (patience=6)
- **Batch size:** 8; **GPU:** NVIDIA A100 (AMP / mixed precision)
- **Data split:** 80/20 train/val from MOT16 training sequences (4253/1063 images)
- **Gradient clipping:** max norm = 1.0

Table 1: Mask R-CNN training progression (A100, 40 epochs, AMP).

Phase	Train Loss	Val Loss
Warmup completed (Epoch 4)	∼0.58	∼0.55
Mid-training (Epoch 20)	∼0.42	∼0.44
Best model (Epoch ∼35)	∼0.36	**0.3783**
Early stop (Epoch 41)	∼0.35	∼0.39

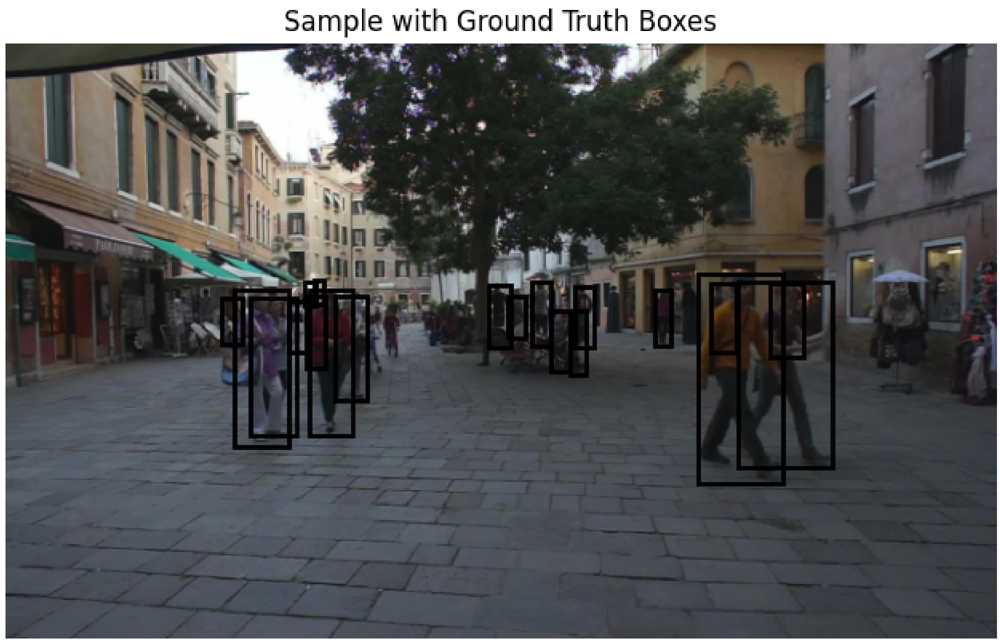

Figure 1: Sample MOT16 training data with ground-truth bounding boxes and instance masks overlaid. Each color represents a distinct person identity.

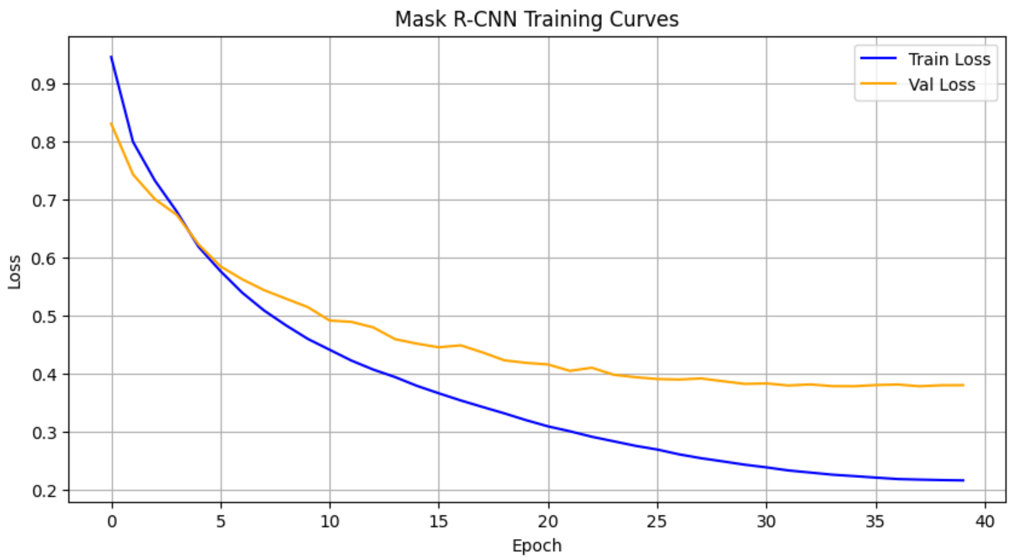

Figure 2: Mask R-CNN training and validation loss curves over training epochs, showing consistent convergence.

Figure 3: Mask R-CNN inference results on a MOT16 test frame, showing detected person bounding boxes with confidence scores and segmentation masks.

2.2 Person Re-Identification — Improved Siamese Network

2.2.1 Architecture

The ReID model is an improved Siamese network built on a **ResNet-50** backbone with the following components:

- **Backbone:** ResNet-50 (up to global average pooling)
- **Embedding head:** Linear(2048→512) → ReLU → Dropout(0.3) → Linear(512→256)
- **BN Neck:** BatchNorm1d(256), following Luo et al. (2019)
- **Classifier:** Linear(256→751, bias=False)

During training, the model outputs L2-normalized embeddings (for triplet loss) and classification logits (for cross-entropy). At inference, only normalized embeddings are returned.

2.2.2 Loss Function

The training objective combines two complementary losses:

$$\mathcal{L} = \mathcal{L}_{\text{triplet}} + w_{\text{CE}} \cdot \mathcal{L}_{\text{CE}}$$

Triplet Loss with margin α operates on raw (pre-BN) embeddings:

$$\mathcal{L}_{\text{triplet}} = \sum_{i}^{N} \max\big(\|f(x_i^a) - f(x_i^p)\|_2^2 - \|f(x_i^a) - f(x_i^n)\|_2^2 + \alpha,\ 0\big)$$

Cross-Entropy Loss with label smoothing operates on post-BN embeddings.

2.2.3 Training Details

- **Dataset:** Market-1501 (751 IDs, 12,936 images)
- **Optimizer:** AdamW (lr=3.5×10^{-4})
- **LR Schedule:** Linear warmup (10 epochs) + Cosine annealing (70 epochs)
- **Total epochs:** 80; **Early stopping:** patience=12
- **Augmentation:** Random flip, random erasing, color jitter
- **Input size:** 256×128, ImageNet normalization

Table 2: ReID training progression (selected epochs).

Epoch	Total Loss	Triplet	CE	LR
1	3.5411	0.6192	5.8039	7.0×10^{-5}
10	2.2576	0.0990	4.3022	3.5×10^{-4}
40	1.8073	0.0177	3.5618	2.5×10^{-4}
60	1.6466	0.0046	3.2840	7.2×10^{-5}
76 (best)	**1.5971**	0.0013	3.1917	4.0×10^{-6}
80	1.5977	0.0017	3.1921	0.0

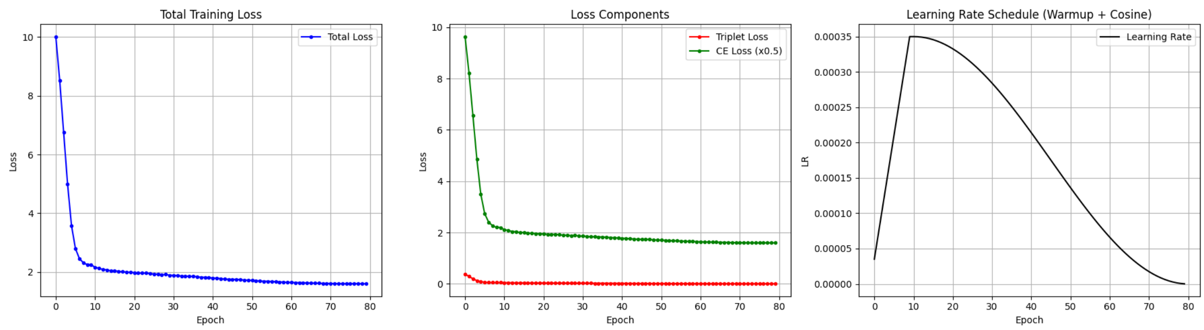

Figure 4: ReID training curves: (Left) Total, Triplet, and CE loss components over 80 epochs. (Right) Learning rate schedule showing linear warmup followed by cosine annealing.

2.2.4 Evaluation Results

Table 3: ReID evaluation on Market-1501 test set.

Metric	Value
mAP	**73.93%**
Rank-1	**89.22%**
Rank-5	96.05%
Rank-10	97.45%

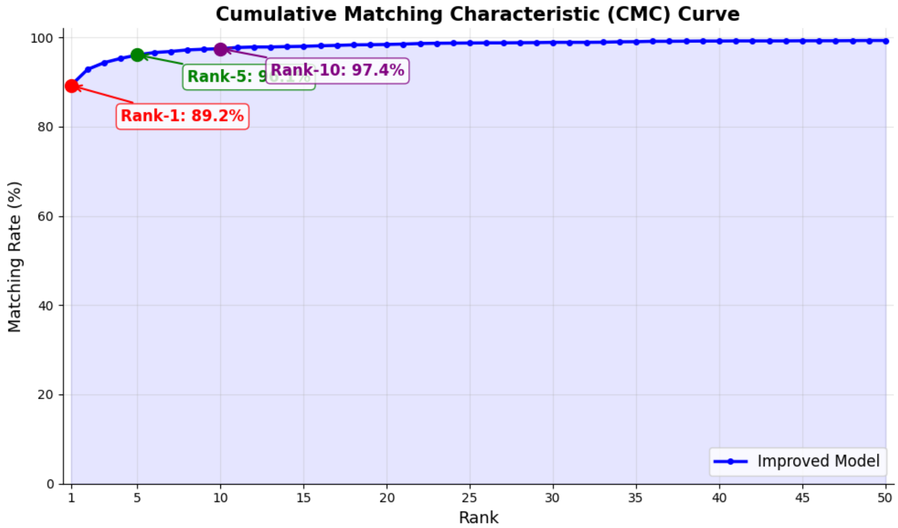

Figure 5: Cumulative Match Characteristic (CMC) curve showing recognition accuracy at different ranks. Rank-1 accuracy reaches 89.22%.

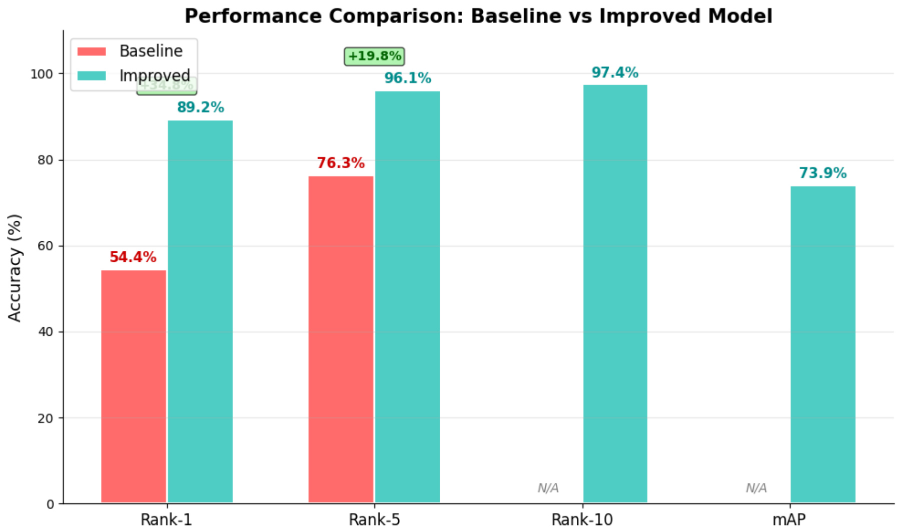

Figure 6: Performance comparison between baseline and improved ReID models across mAP, Rank-1, Rank-5, and Rank-10 metrics.

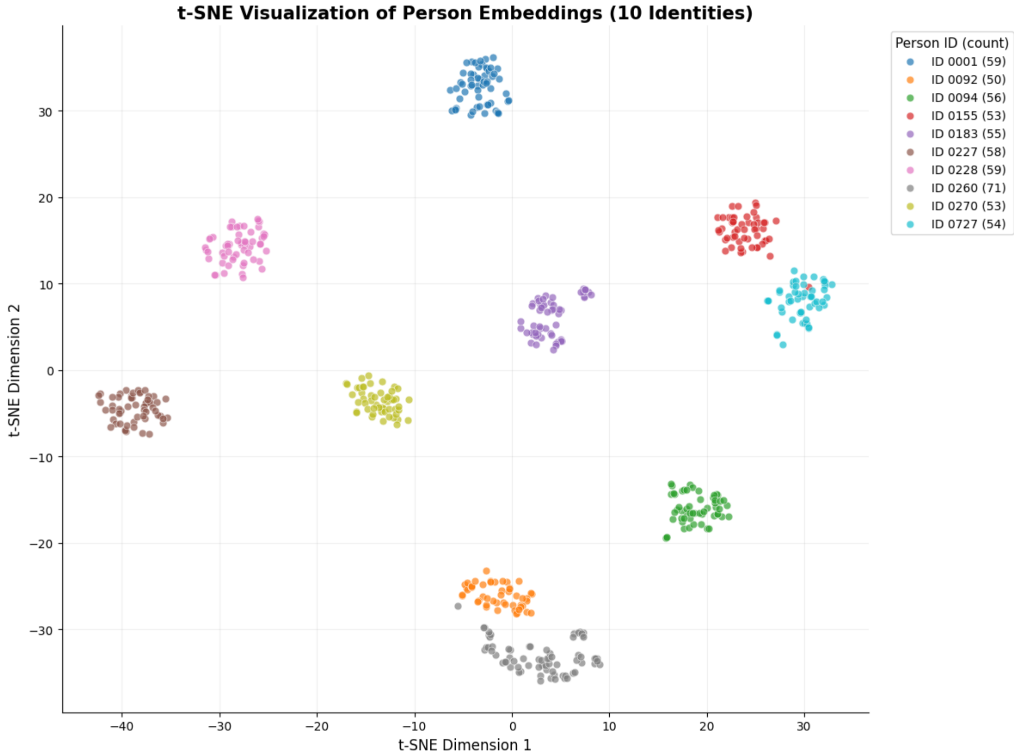

Figure 7: t-SNE visualization of ReID embeddings. Each color represents a distinct person identity, showing clear clustering of same-identity embeddings in the learned feature space.

Figure 8: Top-5 gallery matches for a sample query image. Green borders indicate correct matches (same identity); red borders indicate incorrect matches.

2.3 Multi-Object Tracking — DeepSORT-Style Tracker

2.3.1 Tracking Algorithm

The tracker maintains active tracks, each with a bounding box, appearance embedding, and unique ID. Per-frame association:

1. Run Mask R-CNN to obtain bounding boxes
2. Crop detections, resize to 256×128, extract 256-dim ReID embeddings
3. Build cost matrix: $C_{ij} = w_{\mathrm{IoU}} \cdot (1 - \mathrm{IoU}(t_i, d_j)) + (1 - w_{\mathrm{IoU}}) \cdot d_{\cos}(e_i, e_j), \quad w_{\mathrm{IoU}} = 0.5$
4. Apply Hungarian algorithm for optimal assignment
5. Gate: reject matches with cost > 0.7

6. Update matched embeddings via EMA: $e' = 0.8 \cdot e_{\text{old}} + 0.2 \cdot e_{\text{new}}$
7. Birth: unmatched detections $\rightarrow$ new tracks; Death: missed > 90 frames $\rightarrow$ remove

2.3.2 Results

The pipeline was evaluated on **MOT16-09** (525 frames):

Table 4: Tracking results on MOT16-09.

Metric	Value
Total frames processed	525
Total unique IDs assigned	22
Total track entries	4,480
Active tracks per frame (avg)	$\sim$8–12

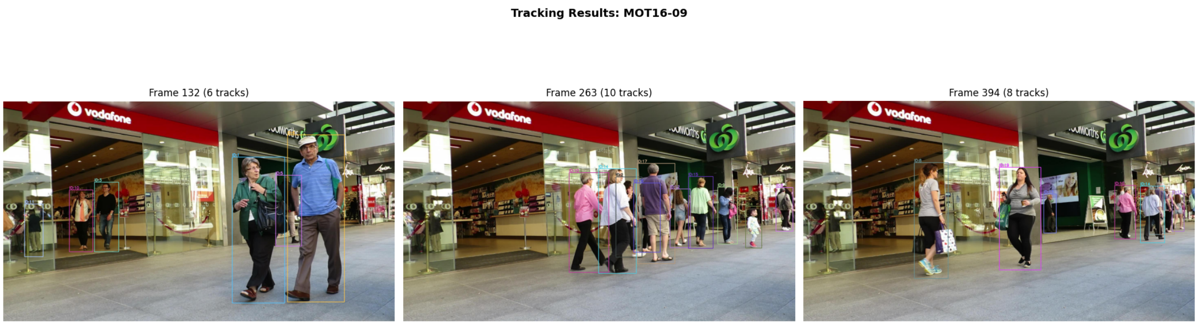

Figure 9: Sample tracking results on MOT16-09 at different time points. Each person is assigned a persistent color-coded ID label maintained across frames by the DeepSORT association pipeline.

3 Challenges Encountered and Solutions

3.1 Limited Training Data for Detection

Challenge: MOT16 contains only 7 training sequences with limited diversity.
Solution: Transfer learning from COCO-pretrained Mask R-CNN with frozen backbone layers. Data augmentation (color jitter, horizontal flip, random cropping) was applied to expand the effective dataset.

3.2 GPU Memory Constraints on Google Colab

Challenge: Training Mask R-CNN with segmentation masks is memory-intensive; Colab sessions are time-limited.
Solution: Reduced input resolution (384/640 vs. default 800/1333), checkpoint saving for session resumption, and early stopping to avoid unnecessary computation.

3.3 Architecture Mismatch Between Detector and ReID

Challenge: The 2-class Mask R-CNN mask predictor weights have different shapes than the default 91-class model, causing loading errors.
Solution: Filtered checkpoint loading that skips mismatched mask predictor keys:

```
filtered = {k: v for k, v in state.items()
            if 'mask_predictor' not in k}
model.load_state_dict(filtered, strict=False)
```

3.4 Identity Switching in Crowded Scenes

Challenge: Similar appearances and overlapping boxes cause incorrect ID assignments.
Solution: Composite cost function balancing IoU and cosine distance (50/50). Exponential moving average embedding updates ($0.8 \cdot e_{\text{old}} + 0.2 \cdot e_{\text{new}}$) provide robustness against transient appearance changes.

3.5 Mask Integration for Tracking Visualization

Challenge: Integrating segmentation masks into tracking visualization while maintaining rendering performance.
Solution: Bounding-box-only tracking first, then mask overlay added in a subsequent iteration. Masks provide visual fidelity but are not used in the tracking cost computation.

4 Self-Evaluation

4.1 Strengths

- **End-to-end pipeline:** Successfully integrates detection, re-identification, and tracking into a cohesive system.
- **Strong ReID:** 73.93% mAP and 89.22% Rank-1 on Market-1501 demonstrate effective appearance modeling.
- **Robust detector:** Best validation loss of 0.3783 achieved via A100-optimized training (AMP, differential LR, warmup+cosine schedule) with accurate person detections and segmentation masks.
- **Modular design:** Each component can be trained, evaluated, and improved independently.

4.2 Limitations

- **No motion model:** No Kalman filter or motion prediction limits occlusion/fast-motion handling.
- **No formal MOT metrics:** MOTA, IDF1, and HOTA were not computed; evaluation relied on qualitative analysis.
- **Limited sequences:** Primarily evaluated on MOT16-09; more challenging sequences would provide fuller assessment.
- **ReID domain gap:** Market-1501 has different characteristics than MOT16, potentially limiting generalization.

4.3 Grade Self-Assessment

Based on the rubric:

- ✓ Detects pedestrians across all frames using fine-tuned Mask R-CNN
- ✓ Assigns bounding boxes with persistent identity labels
- ✓ Tracks objects across the majority of frames
- ✓ Produces annotated video output demonstrating performance

We believe the project merits a grade of **A/B**, as the system demonstrates effective tracking with persistent IDs across most frames, though some ID switches occur in challenging scenarios.

5 Citations

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All source code is available in the accompanying notebooks: `MaskRCNN_A100_Tracking-2.ipynb` (Mask R-CNN), `ReID_Improved.ipynb` (Siamese ReID), and `Tracking_Pipeline.ipynb` (integrated tracking).
GitHub repository: `https://github.com/JoeDoan/-Deep_Learning_Project`